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BAYSIAN INFERENCE / ESTIMATION

In classical point estimation, so far, we assumed that our random sample came from some density $f(x; \theta)$, where $f(x; \theta)$ is assumed to be known. We also assumed that θ was some fixed (unknown) point.

But in some real-world situations, apart from the density $f(x; \theta)$, we have some additional information about θ i.e. θ itself act as random variable. In Bayesian estimation, we combine the additional information about θ and the likelihood function of the r. sample to find the estimator about estimate the about the unknown parameter.

Prior Distribution If θ act as a r.v i.e the unknown parameter θ is the value of some r.v, say Θ , it has a distribution which is called prior distribution.

Posterior Distribution

The condition density of θ given $X_1 = x_1, \dots, x_n$ observed by $f(\theta/x_1, \dots, x_n)$ is called the posterior distribution.

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and for random sampling it is given as

$$f(\theta/x_1, \dots, x_n) = \frac{\prod_{i=1}^n f(x_i/\theta) g(\theta)}{\int \prod_{i=1}^n f(x_i/\theta) g(\theta) \cdot d\theta}$$

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$$f(\theta/\text{data}) = \frac{L(x_1, \dots, x_n) g(\theta)}{\int L(x_1, \dots, x_n) g(\theta) \cdot d\theta}$$

which is a density function of given data.

Posterior Bayes Estimator

Let $x_1, \dots, x_n$ be a random sample from density $f(x; \theta)$, where θ is a value of r.v. θ with known density (prior) $g(\theta)$ of θ .

The posterior Bayes estimator of $T(\theta)$ with respect of $g(\theta)$ is defined as

$$E[T(\theta)/x_1, \dots, x_n]$$

$$= \int T(\theta) f(\theta/x_1, \dots, x_n) d\theta$$

$$= \frac{\int T(\theta) \prod_{i=1}^n f(x_i/\theta) g(\theta) d\theta}{\int \prod_{i=1}^n f(x_i/\theta) g(\theta) \cdot d\theta}$$

(B)

- (i) LHF of $x_1, \dots, x_n \left[\prod_{i=1}^n f(x_i/\theta) \right]$.
- (ii) $g(\theta)$
- (iii) Joint distⁿ $L(\cdot)$ and $g(\theta)$
- (iv) Marginal distⁿ of θ
- (v) Conditional distⁿ = $\frac{JD}{MD}$
- (vi) Expectation of conditional distⁿ is the Bayes estimator.

Derivation of Posterior Distribution.

Posterior Distⁿ: Let θ & x be two random variables their joint distⁿ can be written as.

$$p(\theta, x) = p(x) \cdot p(\theta/x) \quad \text{--- (i)}$$

$$= p(\theta) p(x/\theta) \quad \text{--- (ii)}$$

from eq (i) & (ii)

$$p(x) p(\theta/x) = p(\theta) \cdot p(x/\theta)$$

$$\Rightarrow p(\theta/x) = \frac{p(\theta) p(x/\theta)}{p(x)} \quad \text{--- (iii)}$$

applying $\int$ on both sides eq (iii)

$$\int_{-\infty}^{\infty} p(\theta/x) d\theta = \int_{-\infty}^{\infty} \frac{p(\theta) p(x/\theta)}{p(x)} d\theta$$

$$\begin{aligned} P(A \cap B) &= P(B|A) \cdot P(A) \\ &= P(A) \cdot P(B|A) \\ P(A|B) &= \frac{P(A \cap B)}{P(B)} \end{aligned}$$

$$\begin{aligned} f(x/y) &= \frac{f(x,y)}{f(y)} \\ f(y/x) &= \frac{f(x,y)}{f(x)} \end{aligned}$$

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$$1 = \frac{1}{p(x)} \int_{-\infty}^{\infty} p(\theta) p(x/\theta) d\theta$$

$$p(x) = \int_{-\infty}^{\infty} p(\theta) p(x/\theta) \cdot d\theta$$

$$p(x) = \int p(\theta) p(x/\theta) d\theta$$
$$p(\theta) = \int p(\theta) p(x/\theta) d\theta$$

put in eq (iii)

$$p(\theta/x) = \frac{p(\theta) \cdot p(x/\theta)}{\int_{-\infty}^{\infty} p(\theta) \cdot p(x/\theta) \cdot d\theta}$$

here $p(\theta)$ is known as prior distⁿ of θ ,
consider x as a data (sample observation)
then $p(\theta/x) = p(\theta/\text{data})$ is known as
posterior distⁿ of θ given data and
 $p(x/\theta) = L(x_1, \dots, x_n; \theta)$

so

$$p(\theta/\text{data}) = \frac{p(\theta) L(x_1, \dots, x_n; \theta)}{\int_{-\infty}^{\infty} p(\theta) L(x_1, \dots, x_n; \theta) d\theta}$$

is the posterior distⁿ.

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Assume that the length of Nylon string which can be produced by a machine with no break occurring is an exponential random variable x with parameter λ . Also assume an exponential prior density as λ with parameter $\frac{1}{1000}$. Given a sample of 1 observation on x (and we find $x=1000$). Compute the posterior density for λ and Bayes estimator.

$$f(x) = \lambda e^{-\lambda x} \quad 0 < x < \infty$$

The prior distⁿ of λ is as

$$g(\lambda) = \frac{1}{1000} e^{-\frac{\lambda}{1000}}$$

the LHF is (n=1)

$$\prod_{i=1}^n f(x/\lambda) = \lambda e^{-\lambda x}$$

Now the joint pdf for λ and data is

$$\prod_{i=1}^n f(x) g(\lambda) = \lambda e^{-\lambda x} \frac{1}{1000} e^{-\frac{\lambda}{1000}}$$

$$= \frac{\lambda}{1000} e^{-\lambda(x + \frac{1}{1000})} \quad \text{--- (A)}$$

$$\text{Now } \int_0^{\infty} \prod_{i=1}^n f(x) g(\lambda) d\lambda = \frac{1}{1000} \int_0^{\infty} \lambda e^{-\lambda(x + \frac{1}{1000})} d\lambda$$

$$\text{put } \lambda(x + \frac{1}{1000}) = t$$

$$d\lambda = \frac{1}{x + \frac{1}{1000}} dt \quad \lambda = \left(\frac{1}{x + \frac{1}{1000}} \right) t$$

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$$= \frac{1}{1000} \int_0^{\infty} \frac{t}{x + \frac{1}{1000}} e^{-t} \frac{1}{x + \frac{1}{1000}} dt$$

$$= \frac{1}{1000} \int_0^{\infty} \frac{t}{\left(x + \frac{1}{1000}\right)^2} e^{-t} dt$$

$$= \frac{1}{1000 \left(x + \frac{1}{1000}\right)^2} \int_0^{\infty} t^{2-1} e^{-t} dt$$

$$= \frac{\Gamma 2}{1000 \left(x + \frac{1}{1000}\right)^2} = \frac{1}{1000 \left(x + \frac{1}{1000}\right)^2} \quad \text{--- (B)}$$

Now the posterior distribution density

$$\begin{aligned} f(\lambda/x) &= \frac{A}{B} = \frac{\frac{1}{1000} \lambda e^{-\lambda \left(x + \frac{1}{1000}\right)}}{\frac{1}{1000} \left(x + \frac{1}{1000}\right)^2} \\ &= \left(x + \frac{1}{1000}\right)^2 \lambda e^{-\lambda \left(x + \frac{1}{1000}\right)} \end{aligned}$$

Now the Bayes estimator of λ

$$\begin{aligned} \lambda^* = E(\lambda) &= \int_0^{\infty} \lambda \left(x + \frac{1}{1000}\right)^2 \lambda e^{-\lambda \left(x + \frac{1}{1000}\right)} d\lambda \\ &= \int_0^{\infty} \lambda^2 \left(x + \frac{1}{1000}\right)^2 e^{-\lambda \left(x + \frac{1}{1000}\right)} d\lambda \end{aligned}$$

$$\text{put } \lambda \left(x + \frac{1}{1000}\right) = w$$

$$\lambda = \frac{w}{x + \frac{1}{1000}}$$

$$d\lambda = \frac{dw}{x + \frac{1}{1000}}$$

$$x + \frac{1}{1000}$$

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$$\lambda^* = E(\lambda) = \int_0^{\infty} w^2 \frac{e^{-w}}{\left(\lambda + \frac{1}{1000}\right)} dw$$

$$= \frac{1}{\lambda + \frac{1}{1000}} \int_0^{\infty} w^{3-1} e^{-w} dw$$

$$\lambda^* = \frac{\sqrt{3}}{\lambda + \frac{1}{1000}} = \frac{2}{\lambda + \frac{1}{1000}}$$

Ans.

ple 2 Let $X_1, X_2, \dots, X_n$ denote a random sample from Bernoulli distⁿ. Assume that the prior distⁿ of θ is given by $g(\theta) = I_{(0,1)}(\theta)$ find the posterior Bayes estimator of θ and $J(\theta) = \theta(1-\theta)$.

Sol

The pf of Bernoulli is $f(x) = \theta^x (1-\theta)^{1-x}$.

The LHF is

$$\prod_{i=1}^n f(x_i|\theta) = \theta^{\sum x_i} (1-\theta)^{n - \sum x_i}$$

The prior density $g(\theta) = 1 \quad I_{(0,1)}(\theta)$

Now the joint density of x & θ

$$\prod_{i=1}^n f(x_i|\theta) g(\theta) = \theta^{\sum x_i} (1-\theta)^{n - \sum x_i} \quad \text{--- (A)}$$

$$\int_0^1 \prod_{i=1}^n f(x_i|\theta) g(\theta) d\theta = \int_0^1 \theta^{\sum x_i} (1-\theta)^{n - \sum x_i} d\theta$$

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$$\int_0^1 \prod_{i=1}^n f(x_i/\theta) g(\theta) \cdot d\theta = \int_0^1 \theta^{\sum x_i + 1 - 1} (1-\theta)^{n - \sum x_i + 1 - 1} d\theta$$

$$= \beta(\sum x_i + 1, n - \sum x_i + 1) \quad \text{--- (B)}$$

The posterior distⁿ $p(\theta/\text{data})$ is

$$f(\theta/x) = \frac{A}{\beta} = \frac{\theta^{\sum x_i} (1-\theta)^{n - \sum x_i}}{\beta(\sum x_i + 1, n - \sum x_i + 1)}$$

is the posterior distⁿ.

Now the Bayes estimate for θ is

$$E(\theta) = \theta^* = \int_0^1 \theta \cdot f(\theta/x) \cdot d\theta$$

$$= \frac{1}{\beta(\sum x_i + 1, n - \sum x_i + 1)} \int_0^1 \theta^{\sum x_i + 1} (1-\theta)^{n - \sum x_i + 1 - 1} d\theta$$

$$= \frac{1}{\beta(\sum x_i + 1, n - \sum x_i + 1)} \cdot \beta(\sum x_i + 2, n - \sum x_i + 1)$$

$$= \frac{\frac{\Gamma(\sum x_i + 2) \Gamma(n - \sum x_i + 1)}{\Gamma(\sum x_i + 2 + n - \sum x_i + 1)}}{\frac{\Gamma(\sum x_i + 1) \Gamma(n - \sum x_i + 1)}{\Gamma(\sum x_i + 1 + n - \sum x_i + 1)}}$$

9.

$$E(\theta) = \frac{\overline{\sum x+2} \cdot \overline{n-\sum x+1} \cdot \overline{n+2}}{\overline{n+3} \cdot \overline{\sum x+1} \cdot \overline{n-\sum x+1}}$$

$$= \frac{\cancel{n+2} \cdot (\cancel{\sum x+1}) \cdot \cancel{\sum x+1}}{(\cancel{n+2}) \cdot \cancel{n+2} \cdot (\sum x+1)}$$

$$= \frac{\cancel{n+2} \cdot (\sum x+1) \cdot \cancel{\sum x+1}}{(\cancel{n+2}) \cdot \cancel{n+2} \cdot \cancel{\sum x+1}} = \frac{\sum x+1}{n+2}$$

Now B.E for $T(\theta) = \theta(1-\theta)$

$$E\{T(\theta)\} = \frac{\int_0^1 \theta(1-\theta) \theta^{\sum x} (1-\theta)^{n-\sum x} d\theta}{\beta(\sum x+1, n-\sum x+1)}$$

$$= \frac{1}{\beta(\sum x+1, n-\sum x+1)} \int_0^1 \theta^{\sum x+2-1} (1-\theta)^{n-\sum x+2-1} d\theta$$

$$= \frac{\beta(\sum x+2, n-\sum x+2)}{\beta(\sum x+1, n-\sum x+1)}$$

$$E\{T(\theta)\} = \frac{(\sum x+1)(n-\sum x+1)}{(n+3)(n+2)}$$

ANS.

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Example 3 The RV x follows the exponential distⁿ with parameter λ , we assume the prior density for the unknown λ as

$$f(\lambda) = \frac{1}{m!} \left(\frac{m+1}{\lambda_0} \right)^{m+1} \lambda^m e^{-\frac{(m+1)\lambda}{\lambda_0}} \quad 0 < \lambda < \infty$$

Given R.S of size n obtain Bayes estimate of λ .

Sol

$$f(x/\lambda) = \lambda e^{-\lambda x} \quad 0 \leq x < \infty$$

$$\prod_{i=1}^n f(x_i/\lambda) = \lambda^n e^{-\lambda \sum x_i}$$

~~the the $g(\lambda) = \frac{1}{m!}$~~

Now the joint density of x , λ and $d\lambda$ is

$$\prod_{i=1}^n f(x_i/\lambda) \cdot g(\lambda) = \frac{1}{m!} \left(\frac{m+1}{\lambda_0} \right)^{m+1} \lambda^{m+n} e^{-\lambda \left(\sum x_i + \frac{m+1}{\lambda_0} \right)}$$

$$= K \lambda^{m+n} e^{-\lambda \left(\sum x_i + \frac{m+1}{\lambda_0} \right)} \quad \text{--- (A)}$$

$$\int_0^{\infty} \prod_{i=1}^n f(x_i/\lambda) g(\lambda) d\lambda = K \int_0^{\infty} \lambda^{m+n} e^{-\lambda \left(\sum x_i + \frac{m+1}{\lambda_0} \right)} d\lambda$$

put $\lambda \left(\sum x_i + \frac{m+1}{\lambda_0} \right) = t$

$$\lambda = \frac{t}{\left(\sum x_i + \frac{m+1}{\lambda_0} \right)}$$

$$d\lambda = \frac{dt}{\left(\sum x_i + \frac{m+1}{\lambda_0} \right)}$$

$$= \frac{K}{\left(\sum x + \frac{m+1}{\lambda_0}\right)^{m+n+1}} \int_0^{\infty} t^{m+n} e^{-t} dt$$

$$= \frac{K}{\left(\sum x + \frac{m+1}{\lambda_0}\right)^{m+n+1}} \Gamma(m+n+1) \quad (B)$$

$$f(\lambda/x) = \frac{A}{B} = \frac{K \lambda^{m+n} e^{-\lambda \left(\sum x + \frac{m+1}{\lambda_0}\right)}}{\left(\sum x + \frac{m+1}{\lambda_0}\right)^{m+n+1} \Gamma(m+n+1)}$$

$$f(\lambda/x) = \frac{K}{\left(\sum x + \frac{m+1}{\lambda_0}\right)^{m+n+1} \Gamma(m+n+1)} \lambda^{m+n} e^{-\lambda \left(\sum x + \frac{m+1}{\lambda_0}\right)}$$

$$E(\lambda) = \lambda^* = \int \lambda f(\lambda/x) \cdot d\lambda$$

$$= \frac{\left(\sum x + \frac{m+1}{\lambda_0}\right)^{m+n+1}}{\Gamma(m+n+1)} \int_0^{\infty} \lambda^{m+n+1} e^{-\lambda \left(\sum x + \frac{m+1}{\lambda_0}\right)} d\lambda$$

$$\text{put } \lambda \left(\sum x + \frac{m+1}{\lambda_0}\right) = w$$

$$\lambda = \frac{w}{\left(\sum x + \frac{m+1}{\lambda_0}\right)}$$

$$d\lambda = \frac{dw}{\left(\sum x + \frac{m+1}{\lambda_0}\right)}$$

$$= \frac{\left(\sum x + \frac{m+1}{\lambda_0}\right)^{m+n+1}}{\Gamma(m+n+1)} \int_0^{\infty} \frac{w^{m+n+1}}{\left(\sum x + \frac{m+1}{\lambda_0}\right)^{m+n+1}} \frac{e^{-w}}{\left(\sum x + \frac{m+1}{\lambda_0}\right)} dw$$

$$= \frac{1}{\Gamma(m+n+1)} \cdot \frac{1}{\left(\sum x + \frac{m+1}{\lambda_0}\right)} \int_0^{\infty} w^{m+n+2-1} e^{-w} dw$$

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$$\lambda = E(\lambda) = \frac{(m+n+1) \cancel{m+n+1}}{\cancel{m+n+1} \left(\sum x + \frac{m+1}{\lambda_0} \right)}$$

$$= \frac{(m+n+1)}{\left(\sum x + \frac{m+1}{\lambda_0} \right)}$$

$$= \frac{\lambda_0 (m+n+1)}{\left(\lambda_0 \sum x + m+1 \right)}$$

$$= \frac{\lambda_0 (m+n+1)}{n \bar{x} \lambda_0 + m+1}$$

$$= \lambda_0$$

Example: The r.v. X follows the poisson distⁿ with parameter θ , we assume the prior density for unknown θ as

$$g(\theta) = \frac{1}{m!} \left(\frac{m+1}{\theta_0} \right)^m \theta^{m+1} e^{-\frac{(m+1)\theta}{\theta_0}} \quad 0 < \theta < \infty$$

Given R.S of size 'n' obtain Bayes estimate of θ .

Sol. $f(x|\theta) = \frac{\theta^x e^{-\theta}}{x!} \quad x = 0, 1, 2, \dots$

the likelihood q^n is

$$\prod_{i=1}^n f(x_i|\theta) = \frac{\theta^{\sum x_i} e^{-n\theta}}{\prod_{i=1}^n (x_i!)}$$

the joint distⁿ is

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(13)

$$\begin{aligned}
 \prod_{i=1}^n f(x_i/\theta) g(\theta) &= \frac{\theta^{\sum x} e^{-n\theta}}{\prod_{i=1}^n (x_i)!} \cdot \frac{1}{m!} \left(\frac{m+1}{\theta_0}\right)^{m+1} \theta^m e^{-\theta \frac{(m+1)}{\theta_0}} \\
 &= \frac{\left(\frac{m+1}{\theta_0}\right)^{m+1}}{m! \prod_{i=1}^n (x_i)!} \theta^{\sum x + m} e^{-\theta \left(n + \frac{m+1}{\theta_0}\right)} \\
 &= K \theta^{\sum x + m} e^{-\theta \left(n + \frac{m+1}{\theta_0}\right)} \quad \text{--- (A)}
 \end{aligned}$$

Now integrating to have marginal distⁿ

$$\int_0^{\infty} \prod_{i=1}^n f(x_i/\theta) g(\theta) d\theta = K \int_0^{\infty} \theta^{\sum x + m} e^{-\theta \left(n + \frac{m+1}{\theta_0}\right)} d\theta$$

put $\theta \left(n + \frac{m+1}{\theta_0}\right) = t$

$$\theta = \frac{t}{n + \frac{m+1}{\theta_0}}$$

$$d\theta = \frac{dt}{\left(n + \frac{m+1}{\theta_0}\right)}$$

$$= K \int_0^{\infty} \frac{t^{\sum x + m}}{\left(n + \frac{m+1}{\theta_0}\right)^{\sum x + m + 1}} e^{-t} dt$$

$$= \frac{K}{\left(n + \frac{m+1}{\theta_0}\right)^{\sum x + m + 1}} \int_0^{\infty} t^{\sum x + m + 1 - 1} e^{-t} dt$$

$$= \frac{K}{\left(n + \frac{m+1}{\theta_0}\right)^{\sum x + m + 1}} \left[\sum x + m + 1 \right] \quad \text{--- (B)}$$

$$p(\theta/x) = \frac{A}{B} = \frac{K \theta^{\sum x + m} e^{-\theta \left(n + \frac{m+1}{\theta_0}\right)}}{K \left(n + \frac{m+1}{\theta_0}\right)^{\sum x + m + 1}}$$

$$p(\theta/x) = \frac{\left(n + \frac{m+1}{\theta_0}\right)^{\sum x + m + 1}}{\left[\sum x + m + 1\right]} \theta^{\sum x + m} e^{-\theta \left(n + \frac{m+1}{\theta_0}\right)}$$

Now Bayes's Estimator

$$E(\theta) = \theta^* = \frac{\left(n + \frac{m+1}{\theta_0}\right)^{\sum x + m + 1}}{\left[\sum x + m + 1\right]} \int_0^{\infty} \theta^{\sum x + m + 1} e^{-\theta \left(n + \frac{m+1}{\theta_0}\right)} d\theta$$

put $\theta \left(n + \frac{m+1}{\theta_0}\right) = w$

$$\theta = \frac{w}{\left(n + \frac{m+1}{\theta_0}\right)}$$

$$d\theta = \frac{dw}{\left(n + \frac{m+1}{\theta_0}\right)}$$

$$= \frac{\left(n + \frac{m+1}{\theta_0}\right)^{\sum x + m + 1}}{\left[\sum x + m + 1\right]} \int_0^{\infty} \left(\frac{w}{\left(n + \frac{m+1}{\theta_0}\right)}\right)^{\sum x + m + 1} e^{-w} \frac{dw}{\left(n + \frac{m+1}{\theta_0}\right)}$$

$$= \frac{1}{\left(n + \frac{m+1}{\theta_0}\right) \left[\sum x + m + 1\right]} \int_0^{\infty} w^{\sum x + m + 1} e^{-w} dw$$

$$E(\theta^*) = \frac{\left(\sum x + m + 1\right)}{\left(n + \frac{m+1}{\theta_0}\right)} = \frac{\theta_0 \left(\sum x + m + 1\right)}{\left(n \theta_0 + m + 1\right)}$$

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LOSS FUNCTION

The error between the value of the parameter and the estimator i.e. $\theta - \theta^*$ represented by a function is known as Loss Function.

Let θ^* denote an estimate of $T(\theta)$, the loss function denoted by $L(\theta^*, \theta)$, is defined as a real valued function satisfying the following

- (i) $L(\theta^*, \theta) \geq 0$ for all possible estimates θ^* and θ
- (ii) $L(\theta^*, \theta) = 0$ for $\theta^* = T(\theta)$

The loss function is considered as a measure of error and our object is to select an estimator that makes this error as minimum as possible.

We cannot make the loss small for every possible sample, but we can make the loss small on the ~~over~~ average.

RISK FUNCTION

The Average of loss is called risk function. For a given loss function $L(\theta^*, \theta)$, the Risk function, denoted by $R_{\theta^*}(\theta)$ of an estimator θ^* is defined as

$$R_{\theta^*}(\theta) = E_{\theta}[L(\theta^*, \theta)]$$

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TYPES OF LOSS FUNCTION

(i) If $L(\theta^*, \theta) = (\theta - \theta^*)^2$, then it is known as Squared Error Loss Function (SELF)

(ii) If $L(\theta^*, \theta) = |\theta - \theta^*|$, then it is known as Absolute Error Loss Function (AELF)

→ THE ESTIMATOR OF PARAMETER θ UNDER SQUARED ERROR LOSS FUNCTION (SELF)

Let $L(\theta^*, \theta) = (\theta - \theta^*)^2$, which is SELF taking expectation on both sides.

$$\begin{aligned} E_{\theta/\text{data}} [L(\theta^*, \theta)] &= E[(\theta - \theta^*)^2] = R(\theta^*) \\ &= E(\theta - \theta^*)^2, \text{ where } R(\theta^*) \\ &\text{is known as risk function.} \end{aligned}$$

$$= E[\theta^2 + \theta^{*2} - 2\theta\theta^*]$$

$$R(\theta^*) = E(\theta^2) + \theta^{*2} - 2\theta^*E(\theta)$$

differentiating $R(\theta^*)$ w.r.t θ^*

$$\frac{\partial R(\theta^*)}{\partial \theta^*} = 0 + 2\theta^* - 2E(\theta) = 0$$

$$2E(\theta) - 2\theta^* = 0$$

$$E(\theta) = \theta^*$$

which is known as Baye's estimator of parameter θ

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1. Based on r.s of size 'n' from the $\frac{1}{\theta} I(0, \theta)^x$ density $f(x/\theta) = 1/\theta$, $0 < x < \theta$ with prior distⁿ as $g(\theta) = 1$, $0 < \theta < 1$ obtain Bayes Estimator of θ w.r.t the loss function $L(\theta, t) = \frac{(t - \theta)^2}{\theta^2}$.

2. Given pdf is $f(x/\theta) = 1/\theta$, $0 \leq x \leq \theta$ and the prior distⁿ is

$$g(\theta) = 1, \quad 0 < \theta < 1$$

Since the range of x involves θ we have to merge the range of x and θ and the sample observations y_i

$$0 < x < \theta \quad \text{PUACP} \quad 0 < \theta < 1$$

$$0 < y_1 < y_2 < \dots < y_n < \theta < 1, \text{ hence}$$

the range of θ is

$$y_n \leq \theta \leq 1$$

The posterior distribution is

$$p(\theta/\text{data}) = f(\theta/x_1, x_2, \dots, x_n) = \frac{\prod_{i=1}^n f(x_i/\theta) \cdot g(\theta)}{\int \prod_{i=1}^n f(x_i/\theta) g(\theta) d(\theta)} \quad \text{--- (i)}$$

where

$$\prod_{i=1}^n f(x_i/\theta) = \frac{1}{\theta^n} \quad \text{--- (A)}$$

$$\text{and } \prod_{i=1}^n f(x_i/\theta) g(\theta) = \frac{1}{\theta^n} \quad \text{--- (A)}$$

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$$\begin{aligned}
 \int_{y_n}^1 \prod_{i=1}^n f(x_i/\theta) g(\theta) \cdot d(\theta) &= \int_{y_n}^1 \theta^{-n} d(\theta) \\
 &= \frac{\theta^{-n+1}}{-n+1} \Big|_{y_n}^1 = \frac{1}{1-n} [1 - y_n^{1-n}] \\
 &= \frac{1 - y_n^{1-n}}{1-n} \quad \text{--- (B)}
 \end{aligned}$$

put (A) and (B) in (1)

$$\begin{aligned}
 p(\theta/\text{data}) &= f(\theta/x_i) = \frac{1-n}{\theta^n (1-y_n^{1-n})} = \frac{-(n-1)}{-\theta^n (y_n^{1-n} - 1)} \\
 &= \frac{(n-1)}{\theta^n \left[\frac{1}{y_n^{n-1}} - 1 \right]} = \frac{(n-1) y_n^{n-1}}{\theta^n (1 - y_n^{n-1})}
 \end{aligned}$$

which the posterior distⁿ for θ .

Now, ^{to} find the estimator of θ w.r.t

the loss function $(t-\theta)^2$ we

have to define the ^{Risk} Risk function and minimize it

$$R(t) = \int L(t, \theta) f(\theta/x_1, x_2, \dots, x_n) d(\theta)$$

$$= \int_{y_n}^1 \frac{(t-\theta)^2}{\theta^2} \frac{(n-1) y_n^{n-1}}{\theta^n (1 - y_n^{n-1})} d(\theta)$$

$$= \frac{(n-1) y_n^{n-1}}{(1 - y_n^{n-1})} \int_{y_n}^1 \frac{t^2 + \theta^2 - 2t\theta}{\theta^{n+2}} d(\theta)$$

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$$\frac{\partial R(t)}{\partial t} = \frac{(n-1) y_n^{n-1}}{1-y_n^{n-1}} \int_{y_n}^1 \frac{2t - 2\theta}{\theta^{n+2}} d\theta = 0$$

$$\# \frac{2(n-1) y_n^{n-1}}{1-y_n^{n-1}} \int_{y_n}^1 \left(\frac{t}{\theta^{n+2}} - \frac{2}{\theta^{n+1}} \right) d\theta = 0$$

$$\# \frac{2(n-1) y_n^{n-1}}{1-y_n^{n-1}} \left[t \int_{y_n}^1 \theta^{-(n+2)} d\theta - \int_{y_n}^1 \theta^{-(n+1)} d\theta \right] = 0$$

$$\# \frac{2(n-1) y_n^{n-1}}{1-y_n^{n-1}} \left[t \left. \frac{\theta^{-n-1}}{-n-1} \right|_{y_n}^1 - \left. \frac{\theta^{-n}}{-n} \right|_{y_n}^1 \right] = 0$$

$$\# \frac{2(n-1) y_n^{n-1}}{1-y_n^{n-1}} \left[t \left(\frac{1-y_n^{-(n+1)}}{-n-1} \right) + \left(\frac{1-y_n^{-n}}{-n} \right) \right] = 0$$

$$t \left(\frac{y_n^{-(n+1)} - 1}{n+1} \right) - \frac{y_n^{-n} - 1}{n} = 0$$

$$t = \frac{y_n^{-n} - 1}{n} \cdot \frac{n+1}{y_n^{-(n+1)} - 1}$$

$$= \frac{n+1}{n} \cdot \frac{y_n^{-n} - 1}{y_n^{-(n+1)} - 1} = \frac{n+1}{n} \cdot \frac{1/y_n^n - 1}{1/y_n^{n+1} - 1}$$

$$= \frac{n+1}{n} \frac{(1-y_n^n)/y_n^n}{(1-y_n^{n+1})/y_n^{n+1}} = \frac{n+1}{n} y_n \frac{(1-y_n^n)}{(1-y_n^{n+1})}$$

Example Let x be a r.s from $f(x/\theta) = \theta^x(1-\theta)^{1-x}$, $x=0,1$ where θ is distributed a priori as uniform over $(0,1)$. Find Bayes Estimator of θ w.r.t the loss function: $L(\theta; t) = \frac{(\theta-t)^2}{\theta(1-\theta)}$

Sol.

$$f(x/\theta) = \theta^x(1-\theta)^{1-x}, \quad x=0,1 \quad \text{and}$$

the prior distⁿ is $g(\theta) = 1, \quad 0 < \theta < 1$

the posterior distributed as

$$p(\theta/\text{data}) = f(\theta/x_1, x_2, \dots, x_n) = \frac{\prod_{i=1}^n f(x_i/\theta) g(\theta)}{\int_{i=1}^n \prod_{i=1}^n f(x_i/\theta) g(\theta) \cdot d\theta}$$

$$\text{now } \prod_{i=1}^n f(x_i/\theta) = \theta^{\sum x_i} (1-\theta)^{n-\sum x_i}$$

$$\text{and } \prod_{i=1}^n f(x_i/\theta) g(\theta) = \theta^{\sum x_i} (1-\theta)^{n-\sum x_i} \quad \text{--- (A)}$$

$$\begin{aligned} \int_0^1 \prod_{i=1}^n f(x_i/\theta) g(\theta) d\theta &= \int_0^1 \theta^{\sum x_i} (1-\theta)^{n-\sum x_i} d\theta \\ &= \int_0^1 \theta^{\sum x_i + 1 - 1} (1-\theta)^{n-\sum x_i + 1 - 1} d\theta \end{aligned}$$

$$= B(\sum x_i + 1, n - \sum x_i + 1) \quad \text{--- (B)}$$

putting in equation (i)

(21)

$$p(\theta/\text{data}) = f(\theta/x_1, \dots, x_n) = \frac{\theta^{\sum x_i} (1-\theta)^{n-\sum x_i}}{\beta(\sum x_i + 1, n - \sum x_i + 1)}$$

is the posterior distⁿ.

Now we define the Risk r^n $R(t)$ w.r.t
to loss $r^n (\theta - t)^2 / \theta(1-\theta)$ and minimising
it to have Bayes estimator of θ

$$R(t) = E_{\theta/\text{data}}[L(\theta, t)] = \int L(\theta, t) f(\theta/x_1, \dots, x_n) d(\theta)$$

$$= \frac{1}{\beta(\sum x_i + 1, n - \sum x_i + 1)} \int \frac{(\theta - t)^2}{\theta(1-\theta)} \theta^{\sum x_i} (1-\theta)^{n-\sum x_i} d(\theta)$$

$$= \frac{1}{\beta(\sum x_i + 1, n - \sum x_i + 1)} \int_0^1 (\theta - t)^2 \theta^{\sum x_i - 1} (1-\theta)^{n-\sum x_i - 1} d(\theta)$$

$$R(t) = \frac{1}{\beta(\sum x_i + 1, n - \sum x_i + 1)} \int_0^1 (\theta^2 + t^2 - 2\theta t) \theta^{\sum x_i - 1} (1-\theta)^{n-\sum x_i - 1} d(\theta)$$

$$\frac{\partial R(t)}{\partial t} = \frac{1}{\beta(\sum x_i + 1, n - \sum x_i + 1)} \int_0^1 (2t - 2\theta) \theta^{\sum x_i - 1} (1-\theta)^{n-\sum x_i - 1} d(\theta) = 0$$

$$\frac{1}{\beta(\sum x_i + 1, n - \sum x_i + 1)} \left[2t \int_0^1 \theta^{\sum x_i - 1} (1-\theta)^{n-\sum x_i - 1} d\theta - 2 \int_0^1 \theta^{\sum x_i} (1-\theta)^{n-\sum x_i - 1} d\theta \right] = 0$$

(17)

$$\frac{1}{\beta(\sum x_i + 1, n - \sum x_i + 1)} \left[2t \beta(\sum x_i, n - \sum x_i) - 2 \beta(\sum x_i + 1, n - \sum x_i) \right]$$

$$\frac{2t \beta(\sum x_i, n - \sum x_i)}{\beta(\sum x_i + 1, n - \sum x_i + 1)} - \frac{2 \beta(\sum x_i + 1, n - \sum x_i)}{\beta(\sum x_i + 1, n - \sum x_i + 1)}$$

$$2t \frac{\frac{\overline{\sum x_i} \overline{n - \sum x_i}}{\overline{\sum x_i + 1} \overline{n - \sum x_i + 1}}}{\frac{\overline{\sum x_i} \overline{n - \sum x_i}}{\overline{\sum x_i + 1} \overline{n - \sum x_i + 1}}} = 2 \frac{\frac{\overline{\sum x_i + 1} \overline{n - \sum x_i}}{\overline{\sum x_i + 1} \overline{n - \sum x_i + 1}}}{\frac{\overline{\sum x_i + 1} \overline{n - \sum x_i + 1}}{\overline{\sum x_i + 1} \overline{n - \sum x_i + 1}}}$$

$$t \frac{\frac{\overline{\sum x_i} \overline{n - \sum x_i}}{n}}{\frac{\overline{\sum x_i} \overline{n - \sum x_i}}{n}} = \frac{\frac{\overline{\sum x_i + 1} \overline{n - \sum x_i}}{n+1}}{\frac{\overline{\sum x_i + 1} \overline{n - \sum x_i + 1}}{n+1}}$$

$$t \frac{n(n+1)}{\sum x_i (n - \sum x_i)} = \frac{(n+1)}{(n - \sum x_i)}$$

$$t = \frac{\cancel{n+1}}{(n - \sum x_i)} \frac{\sum x_i (n - \sum x_i)}{\cancel{n(n+1)}}$$

$$t = \frac{\sum x_i}{n} = \bar{x}$$

hence $\bar{x}$ is the Baye's estimator for θ .

(23)

(Y)



x of size n , is taken from Bernoulli distⁿ having p.d.f p.d:

$$f(x) = \alpha^x (1-\alpha)^{1-x}$$

If the prior distⁿ of α is:

$$p(\alpha) = \frac{1}{\beta(a,b)} \alpha^{a-1} (1-\alpha)^{b-1} \quad a, b > 0, 0 < \alpha < 1$$

Then show that the mean of the posterior distⁿ

$$E(\alpha/\text{data}) = \gamma_n \bar{x} + (1 - \gamma_n) \alpha_0$$

and hence $\gamma_n \rightarrow 1$ as $n \rightarrow \infty$

Sol

$$f(x) = \alpha^x (1-\alpha)^{1-x}$$

$$(\cdot) = \alpha^{\sum x} (1-\alpha)^{n-\sum x} = \prod f(x; \alpha)$$

The joint distⁿ of x & α is

$$\prod f(x; \alpha) g(\alpha) = \frac{1}{\beta(a,b)} \alpha^{a-1} (1-\alpha)^{b-1} \alpha^{\sum x} (1-\alpha)^{n-\sum x}$$

$$\prod f(x; \alpha) = \frac{1}{\beta(a,b)} \alpha^{a+\sum x-1} (1-\alpha)^{b+n-\sum x-1} \quad (A)$$

the marginal distⁿ of α is

$$\int \prod f(x; \alpha) g(\alpha) d(\alpha) = \int \frac{1}{\beta(b+a)} \alpha^{a+\sum x-1} (1-\alpha)^{b+n-\sum x-1} d\alpha \quad (B)$$

The posterior distⁿ is

$$p(\theta/\text{data}) = \frac{\theta^{a+\sum x_i - 1} (1-\theta)^{b+n-\sum x_i - 1}}{\int_0^1 \theta^{a+\sum x_i - 1} (1-\theta)^{b+n-\sum x_i - 1} d\theta}$$

$$= \frac{\theta^{\alpha-1} (1-\theta)^{\beta-1}}{\int_0^1 \theta^{\alpha-1} (1-\theta)^{\beta-1} d\theta}$$

Where $\alpha = a + \sum x_i$
 $\beta = b + n - \sum x_i$

As we know

$$f(x) = \frac{1}{\beta(\alpha, \beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

$$\int_0^1 f(x) dx = \frac{1}{\beta(\alpha, \beta)} \int_0^1 \theta^{\alpha-1} (1-\theta)^{\beta-1} d\theta$$

$$1 =$$

$$\beta(\alpha, \beta) = \int_0^1 \theta^{\alpha-1} (1-\theta)^{\beta-1} d\theta$$

putting this result in (i)

$$p(\theta/\text{data}) = \frac{1}{\beta(\alpha, \beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

which is a beta distⁿ with parameters α and β where α & β are defined above

(25)



We know the mean of the beta distⁿ is

$$E(X) = \text{Mean} = \frac{\alpha}{\alpha + \beta}$$

then the mean of the posterior distⁿ is

$$E(\theta/\text{data}) = \frac{\alpha + \sum x_i}{\alpha + \sum x_i + b + n - \sum x_i} = \frac{a + \sum x_i}{a + b + n}$$

$$= \frac{a}{a + b + n} + \frac{\sum x_i}{a + b + n}$$

$$= \frac{n \bar{x}}{a + b + n} + \frac{a}{a + b + n}$$

$$= \frac{n \bar{x}}{a + b + n} + \frac{a}{a + b + n} \cdot \frac{a + b}{a + b}$$

$$= \bar{x} \frac{n}{a + b + n} + \frac{a + b}{a + b + n} \cdot \frac{a}{a + b}$$

$$\text{Let } M_0 = \frac{a}{a + b}$$

$$= \bar{x} \frac{n}{a + b + n} + \frac{a + b + n - n}{a + b + n} \cdot M_0$$

$$= \bar{x} \frac{n}{a + b + n} + \left[\frac{a + b + n}{a + b + n} - \frac{n}{a + b + n} \right] M_0$$

$$E(\theta/\text{data}) = \bar{x} \frac{n}{a + b + n} + \left[1 - \frac{n}{a + b + n} \right] M_0$$

$$\text{put } \frac{a}{a + b + n} = \gamma_n$$



$$E(O/data) = \gamma_n \bar{x} + (1 - \gamma_n) \mu_0$$

which is the required result.

Now we have to show that $\gamma_n \rightarrow 1$ as $n \rightarrow \infty$

$$\begin{aligned} \gamma_n &= \frac{n}{a+b+n} \\ &= \frac{n/n}{a+b+n/n} = \frac{1}{\frac{a+b}{n} + 1} \end{aligned}$$

$$\gamma_n = \frac{1}{1 + \frac{a+b}{n}}$$

$$\lim_{n \rightarrow \infty} \gamma_n = \lim_{n \rightarrow \infty} \left(\frac{1}{1 + \frac{a+b}{n}} \right) = \frac{1}{1+0} = \frac{1}{1} = 1$$

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If a r.v of size n , follows poisson distⁿ

$$f(x) = \frac{e^{-\theta} \theta^x}{x!} \quad \text{and the}$$

prior distⁿ of θ is a gamma distⁿ with parameter a and b as:

$$g(\theta) = \frac{b^a}{\Gamma(a)} \theta^{a-1} e^{-b\theta}, \quad \theta > 0, \quad a, b > 0$$

then show that the mean of the posterior distⁿ is

$$Y_n \bar{X} + (1 - Y_n) \frac{b}{a}$$

and also show $Y_n \rightarrow 1$ if $n \rightarrow \infty$.

Sol. The posterior distⁿ of θ given the data is

$$\begin{aligned} p(\theta/\text{data}) &= \frac{g(\theta) L(x_1, \dots, x_n; \theta)}{\int_0^\infty g(\theta) L(x_1, \dots, x_n; \theta) \cdot d\theta} \\ &= \frac{\frac{b^a}{\Gamma(a)} \theta^{a-1} e^{-b\theta} e^{-n\theta} \prod_{i=1}^n \frac{e^{x_i} \theta^{x_i}}{x_i!}}{\int_0^\infty \frac{1}{\prod(x_i)!} \frac{b^a}{\Gamma(a)} \theta^{a-1} e^{-b\theta} e^{-n\theta} \theta^{x_i} d\theta} \\ p(\theta/\text{data}) &= \frac{\theta^{a+\sum x_i - 1} e^{-\theta(b+n)}}{\int_0^\infty \theta^{a+\sum x_i - 1} e^{-\theta(b+n)} d\theta} \quad \text{--- (1)} \end{aligned}$$

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$$= \frac{\alpha^{\alpha-1} e^{-\beta\alpha}}{\int_0^{\infty} \alpha^{\alpha-1} e^{-\beta\alpha} d\alpha} \quad (i)$$

where $\alpha = a + \sum x_i$, $\beta = b + n$.

As we know

$$\int_0^{\infty} \alpha^{\alpha-1} e^{-\beta\alpha} d\alpha = \int_0^{\infty} \left(\frac{y}{\beta}\right)^{\alpha-1} e^{-y} \frac{dy}{\beta}$$

by putting $y = \beta\alpha$

$$\Rightarrow \alpha = \frac{y}{\beta}$$

$$d\alpha = \frac{dy}{\beta}$$

$$= \frac{1}{\beta^{\alpha}} \int_0^{\infty} y^{\alpha-1} e^{-y} dy$$

$$= \frac{\Gamma\alpha}{\beta^{\alpha}} \quad \text{put in eq (i)}$$

$$= \frac{\beta^{\alpha}}{\Gamma\alpha} \alpha^{\alpha-1} e^{-\beta\alpha}$$

which is a Gamma distⁿ with parameters α and β , where α and β are

defined above.

Now mean of Posterior distⁿ

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As we know the mean of Gamma distⁿ is

α/β

$$\text{Now } E(\theta/\text{data}) = \frac{a + \sum x_i}{b+n} = \frac{a}{b+n} + \frac{\sum x_i}{b+n}$$

$$= \frac{n\bar{x}}{b+n} + \frac{a}{b+n} \cdot \frac{b}{b}$$

$$= \bar{x} \left(\frac{n}{b+n} \right) + \frac{b}{b+n} \cdot \frac{a}{b}$$

put $a/b = \mu_0$

$$= \bar{x} \left(\frac{n}{b+n} \right) + \frac{b+n-n}{b+n} \mu_0$$

$$E(\theta/\text{data}) = \bar{x} \left(\frac{n}{b+n} \right) + \left(1 - \frac{n}{b+n} \right) \mu_0$$

$$\text{put } \gamma_n = \frac{n}{b+n}$$

$$= \gamma_n \bar{x} + (1 - \gamma_n) \mu_0$$

the required result

Now $\gamma_n \rightarrow 1$ as $n \rightarrow \infty$

$$\gamma_n = \frac{n}{b+n} = \frac{1}{\frac{b}{n} + 1}$$

$$\lim_{n \rightarrow \infty} \gamma_n = \lim_{n \rightarrow \infty} \left(\frac{1}{\frac{b}{n} + 1} \right) = \frac{1}{0+1} = \frac{1}{1} = 1$$

(30)

(8)

A r. sample of $n=1000$ gave $\sum x_i = 710$
from a Bernoulli distⁿ as:

$$f(x) = \theta^x (1-\theta)^{1-x} \quad \begin{matrix} x=0,1 \\ 0 < \theta < 1 \end{matrix}$$

A Statistician 'A' used prior density
of θ as $p(\theta) = 2\theta \quad 0 < \theta < 1$
and another Statistician 'B' used
prior density of θ as:

$$p_B(\theta) = 4\theta^3, \quad 0 < \theta < 1$$

Show which Statistician is more
precise in estimating Bayes estimator?

Sol

$$f(x) = \theta^x (1-\theta)^{1-x} \quad \text{the LHF is}$$

$$L(x_1, \dots, x_n) = \theta^{\sum x_i} (1-\theta)^{n - \sum x_i}$$

the prior distⁿ of θ used by Statistician 'A' is

$$p_A(\theta) = 2\theta$$

the Posterior distⁿ will be

$$p_n(\theta/\text{data}) = \frac{2\theta \theta^{\sum x_i} (1-\theta)^{n - \sum x_i}}{\int_0^1 2\theta \theta^{\sum x_i} (1-\theta)^{n - \sum x_i} d\theta}$$

$$p_n(\theta/\text{data}) = \frac{\theta^{1 + \sum x_i} (1-\theta)^{n - \sum x_i}}{\int_0^1 \theta^{1 + \sum x_i} (1-\theta)^{n - \sum x_i} d\theta}$$

$$p_A(\theta/\text{data}) = \frac{\theta^{2+\sum x_i - 1} (1-\theta)^{1+n-\sum x_i - 1}}{\int_0^1 \theta^{2+\sum x_i - 1} (1-\theta)^{1+n-\sum x_i - 1} d\theta}$$

$$= \frac{\theta^{\alpha-1} (1-\theta)^{\beta-1}}{\int_0^1 \theta^{\alpha-1} (1-\theta)^{\beta-1} d\theta} \quad \text{--- (i)}$$

where

$$\alpha = 2 + \sum x_i$$

$$\beta = 1 + n - \sum x_i$$

Now $f(x) = \frac{1}{\beta(\alpha, \beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$

$$\int_0^1 f(x) \cdot d\theta = \frac{1}{\beta(\alpha, \beta)} \int_0^1 \theta^{\alpha-1} (1-\theta)^{\beta-1} d\theta$$

$$\beta(\alpha, \beta) = \int_0^1 \theta^{\alpha-1} (1-\theta)^{\beta-1} d\theta$$

put in (i)

$$p_A(\theta/\text{data}) = \frac{1}{\beta(\alpha, \beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

which is a beta distⁿ with parameter α and β where α & β are defined above

Now mean of beta distⁿ is:

$$E_A(\theta/\text{data}) = \frac{\alpha}{\alpha + \beta} = \frac{2 + \sum x_i}{2 + \sum x_i + 1 + n - \sum x_i} = \frac{2 + \sum x_i}{3 + n}$$

(32)

(10)

$$E_A(\theta/\text{data}) = \frac{2+710}{3+1000} = \frac{712}{1003} = 0.7098764$$

$$V_A(\theta/\text{data}) = \frac{\alpha\beta}{(\alpha+\beta+1)(\alpha+\beta)^2}$$

$$= \frac{(291)(712)}{(1004)(1003)^2} = 0.0002051$$

$$\alpha = 2 + \sum x_i = 712$$

$$\beta = 1 + n - \sum x_i = 1 + 1000 - 710 = 291$$

$$\frac{712}{1003}$$

Now the precision = $\frac{1}{\text{Var}}$

$$= \frac{1}{0.0002051} = 4874.8650$$

Now STATISTICIAN B

The posterior distⁿ of θ/data is

$$p_B(\theta/\text{data}) = \frac{4\theta^3 \theta^{\sum x_i} (1-\theta)^{n-\sum x_i}}{\int_0^1 4\theta^3 \theta^{\sum x_i} (1-\theta)^{n-\sum x_i} d\theta}$$

$$= \frac{\theta^{3+\sum x_i} (1-\theta)^{n-\sum x_i}}{\int_0^1 \theta^{3+\sum x_i} (1-\theta)^{n-\sum x_i} d\theta}$$

$$= \frac{\theta^{4+\sum x_i-1} (1-\theta)^{1+n-\sum x_i-1}}{\int_0^1 \theta^{4+\sum x_i-1} (1-\theta)^{1+n-\sum x_i-1} d\theta}$$

$$= \frac{\theta^{\alpha-1} (1-\theta)^{\beta-1}}{\int_0^1 \theta^{\alpha-1} (1-\theta)^{\beta-1} d\theta} \quad \text{--- (1)}$$

where $\alpha = 4 + \sum x_i$; $\beta = 1 + n - \sum x_i$

This is again beta distⁿ with parameter α and β where α & β are defined above.

$$\alpha = 4 + \sum x_i = 4 + 710 = 714$$

$$\beta = 1 + n - \sum x_i = 1 + 1000 - 710 = 291$$

$$\text{Mean} = \frac{\alpha}{\alpha + \beta} (\theta / \text{data})$$

$$= \frac{\alpha}{\alpha + \beta} = \frac{714}{714 + 291}$$

$$= 0.7104478$$

$$V_B (\theta / \text{data}) = \frac{\alpha \beta}{(\alpha + \beta + 1)(\alpha + \beta)^2} = \frac{(714)(291)}{(1006)(1005)^2}$$

$$= 0.0002045$$

$$\text{Precision} = \frac{1}{\text{Var}} = \frac{1}{0.0002045} = 4880.338$$

hence Statistician B' is more precise

[Signature]